

EXP NO:7

Introduction to Cloud Computing and AWS/Azure/GCP Console Creating virtual machines, basic compute and storage services.

Aim: To Establishing an AWS Account, Launching an EC2 instance and connecting to it

Objective:

- To create an AWS account.
- To launch an EC2 (Elastic Compute Cloud) instance using the AWS Management Console.
- To connect to the instance via SSH or EC2 Instance Connect.

Overview of Major Cloud Platforms:

Amazon Web Services (AWS)

- World's largest cloud provider
- Services: EC2, S3, RDS, Lambda
- Used by startups to large enterprises

Microsoft Azure:

- Strong integration with Microsoft tools
- Services: Azure VM, Blob Storage, Azure SQL
- Popular in enterprise and academic environments

Google Cloud Platform (GCP)

- Leader in Data & AI: Known for high-performance infrastructure and cutting-edge data analytics.
- Services: * Compute Engine: Scalable virtual machines (Equivalent to EC2/Azure VM).
 - Cloud Storage: Unified object storage (Equivalent to S3/Blob Storage).
 - Cloud SQL: Fully managed relational databases (Equivalent to RDS/Azure SQL).
 - BigQuery: Enterprise data warehouse for large-scale analytics.
 - Vertex AI: Unified platform for machine learning and generative AI.

Pre-Requisites:

1. Internet connection.
2. A valid email ID and credit/debit card for AWS account creation (AWS Free Tier available).
3. Basic knowledge of Linux commands (if connecting to a Linux instance).
4. AWS Free Tier eligibility preferred.

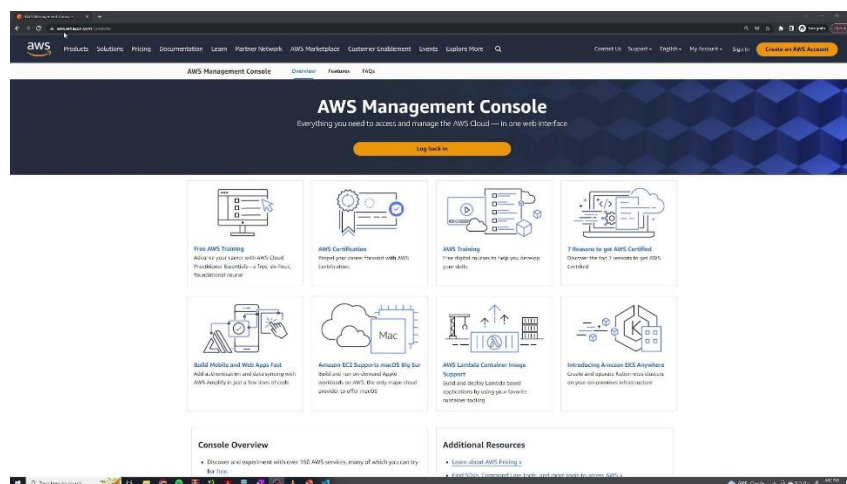
Lab Requirements:

- AWS Management Console (<https://aws.amazon.com/console/>)
- Browser: Chrome/Firefox/Edge
- SSH client (like PuTTY or Linux/Mac terminal) for remote connection

Procedure:

Part A: Create an AWS Account

1. Open AWS Console (<https://aws.amazon.com/>).



2. Click on Create an AWS Account.



Explore Free Tier products with a new AWS account.

To learn more, visit aws.amazon.com/free.



Sign up for AWS

Root user email address
Used for account recovery and some administrative functions

AWS account name
Choose a name for your account. You can change this name in your account settings after you sign up.

Verify email address

OR

Sign in to an existing AWS account

3. Provide: Email ID, Password, AWS Account Name.

Sign up for AWS

Email address
You will use this email address to sign in to your new AWS account.

rajurims86@gmail.com

Password
.....

Confirm password
.....

AWS account name
Choose a name for your account. You can change this name in your account settings after you sign up.

Azurelessons

Continue (step 1 of 5)

[Sign in to an existing AWS account](#)

4. Choose Personal/Professional account type.

Free Tier offers

All AWS accounts can explore 3 different types of free offers, depending on the product used.



Always free
Never expires



12 months free
Start from initial sign-up date



Trials
Start from service activation date

Sign up for AWS

Contact Information

How do you plan to use AWS?

- ☐ **Business** for your work, school, or organization
- ☐ **Personal** for your own projects

Who should we contact about this account?

Full Name

Phone Number

+1 222-333-4444

Country or Region

United States

Address

Apartment, suite, unit, building, floor, etc.

City

State, Province, or Region

Postal Code

☐ I have read and agree to the terms of the [AWS Customer Agreement](#).

Continue (step 2 of 5)

5. Enter Contact information and Payment details.



English ▾

Secure verification

We will not charge you for usage below AWS Free Tier limits. We may temporarily hold up to \$1 USD (or an equivalent amount in local currency) as a pending transaction for 3-5 days to verify your identity.



Sign up for AWS

Billing Information

Credit or Debit card number



AWS accepts all major credit and debit cards. To learn more about payment options, review our FAQ.

Expiration date

Cardholder's name

Billing address

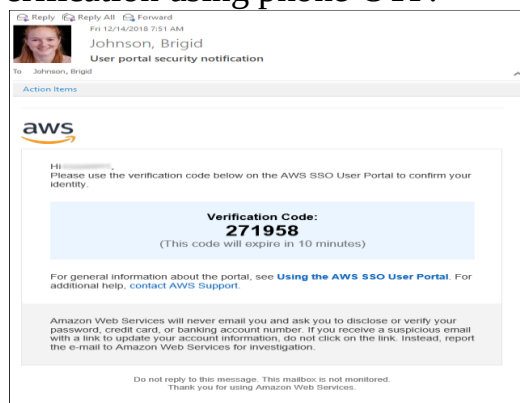
☒ Use my contact address

☐ Use a new address

[Verify and Continue \(step 3 of 5\)](#)

You might be redirected to your bank's website to authorize the verification charge.

6. Complete identity verification using phone OTP.



7. Select Basic (Free Tier) support plan.

AWS Free Tier

Gain free, hands-on experience with the AWS platform, products, and services.

[Create a Free Account](#)

Types of offers

Explore more than 60 products and start building on AWS using the free tier. Three different types of free offers are available depending on the product used. See below for details on each product.

Always free	12 months free	Trials
These free tier offers do not expire and are available to all AWS customers.	Enjoy these offers for 12 months following your initial sign-up date to AWS.	Short-term free trial offers are available through many different software solutions.

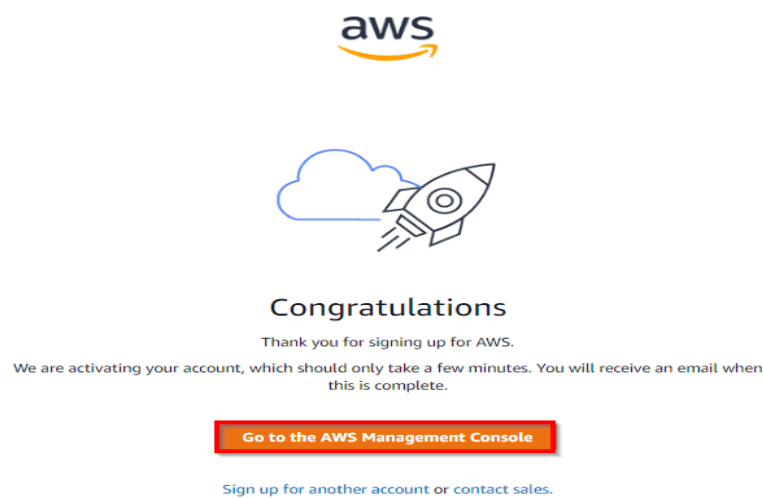
Free Tier details

8. Sign in to AWS Management Console using Root User (email ID).



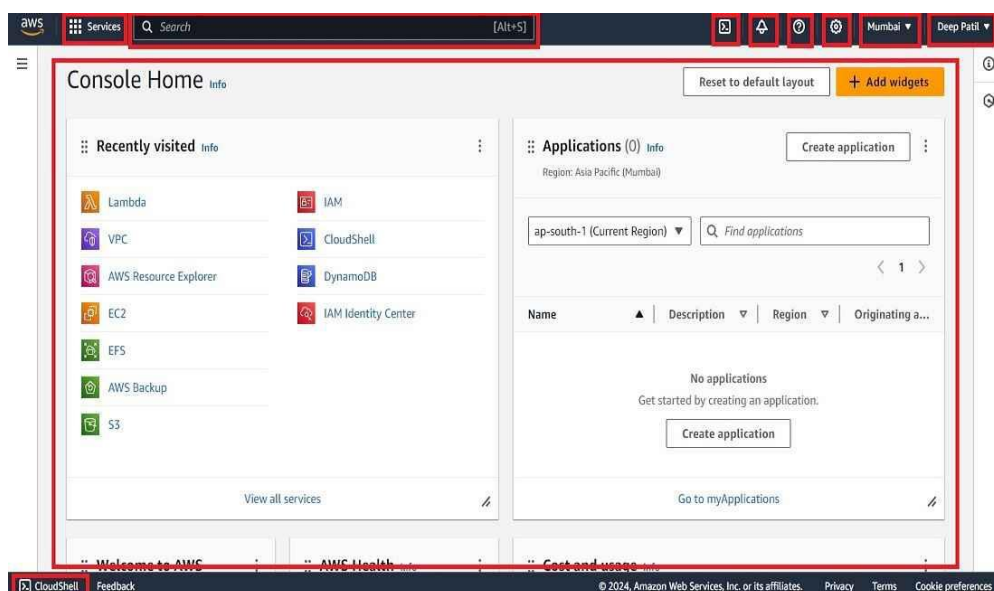
The image shows the AWS Root user sign-in page. At the top is the AWS logo. Below it is the heading "Root user sign in" with an information icon. There are two input fields: "Email:" and "Password:". To the right of the password field is a link "Forgot password?". Below the password field is a blue "Sign in" button. At the bottom, there are two links: "Sign in to a different account" and "Create a new AWS account".

9. Successfully authenticated using the root or IAM user credentials, resulting in redirection to the **AWS Management Console home page**.

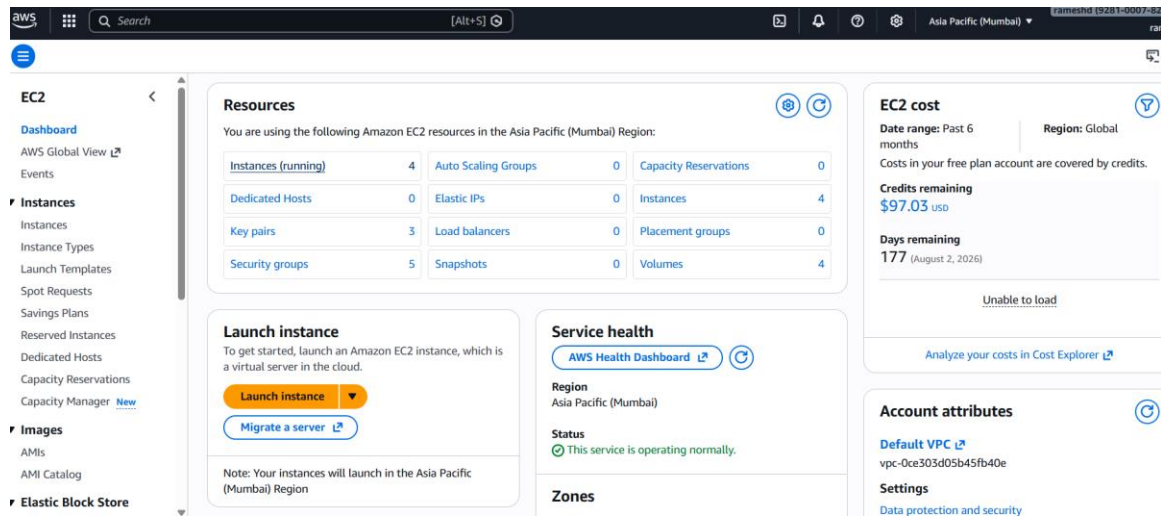


Part B: Launch an EC2 Instance

1. Sign in to AWS Console ☐ Search for EC2 service.

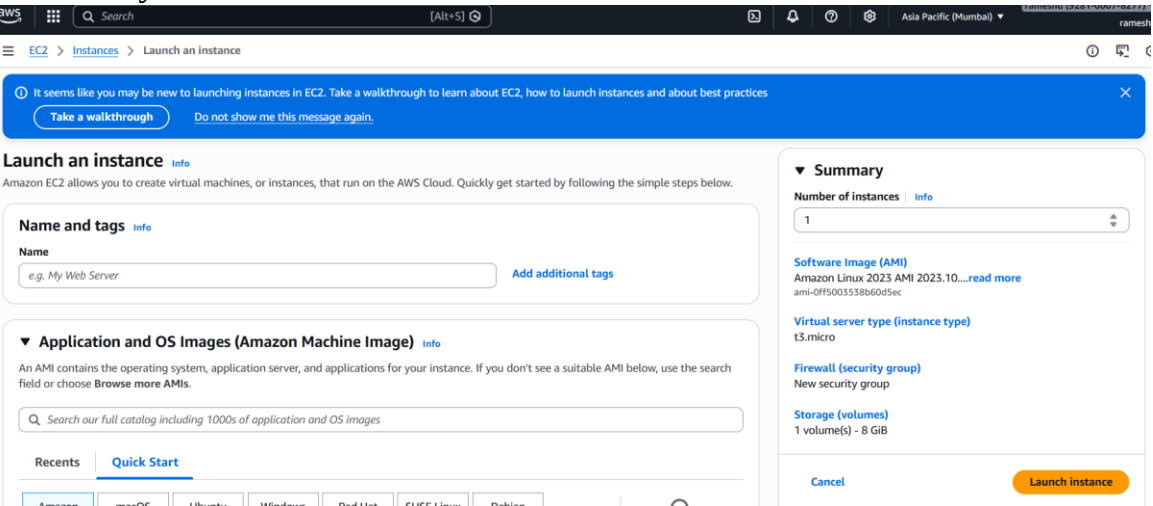


2. Click Launch Instance.



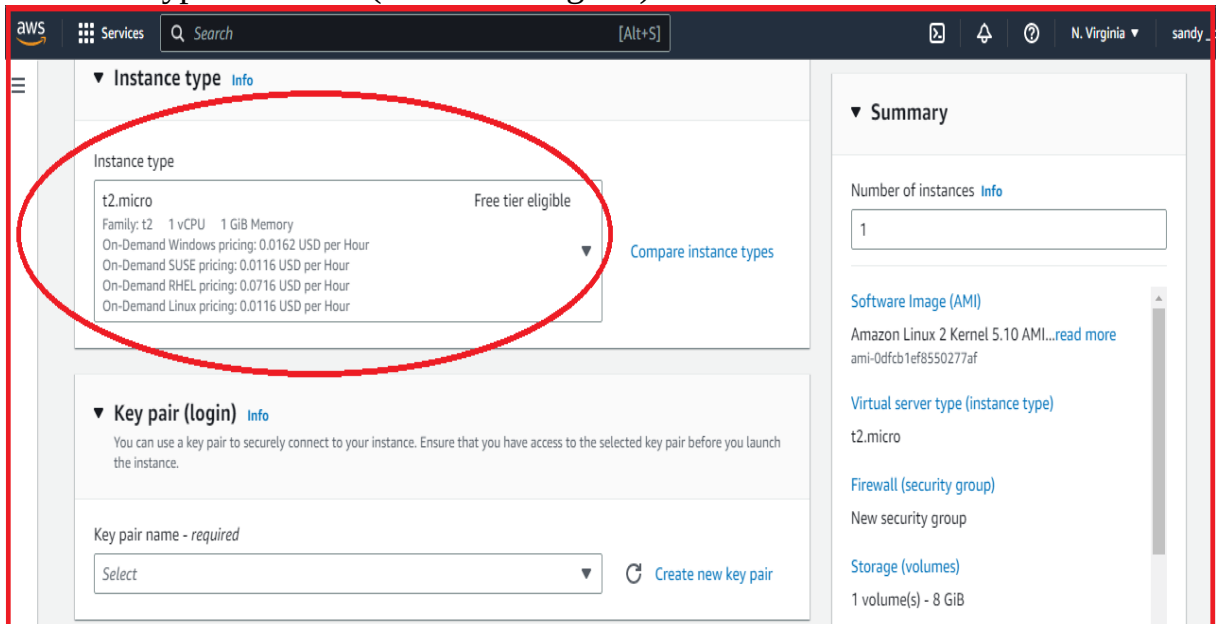
3. Configure:

- Name: MyFirstInstance



- AMI: Amazon Linux 2 AMI (Free Tier eligible)

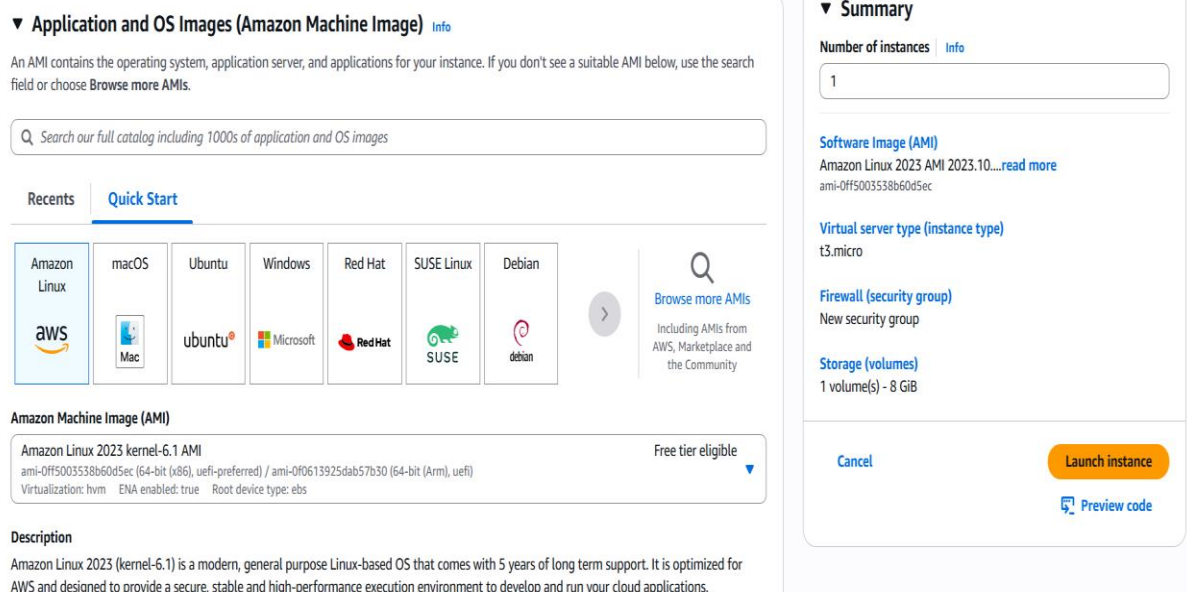
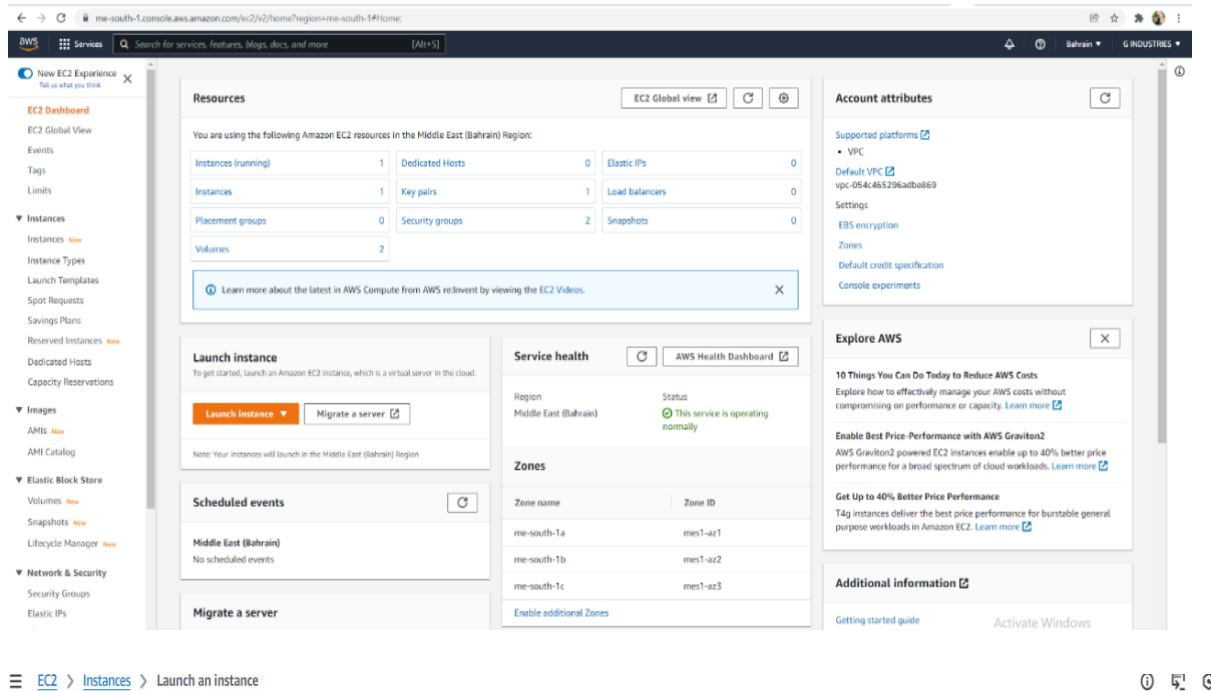
- Instance Type: t2.micro (Free Tier eligible)



- Key Pair: Create new key pair (download .pem file)

- Network Settings: Allow SSH (22), HTTP (80 optional)

- Storage: Default (8GB SSD)



4. Click Launch Instance.
5. After a few minutes, instance state ☐ Running.

Part C: Connect to EC2 Instance

Method 1: Using EC2 Instance

Connect

1. Go to EC2 ☐ Instances ☐ Select instance.
2. Click Connect ☐ Choose EC2 Instance Connect ☐ Click Connect.
3. A browser-based terminal will open with Linux shell access.

The screenshot shows the AWS Management Console for the 'Asia Pacific (Mumbai)' region. The 'EC2' section is selected, and the 'Instances' page is displayed. A table lists the instances, with 'My-Demo-1237' (ID: i-0aa561622f6e5157c) selected. The details for this instance are shown on the right, including its state (Running), type (t3.micro), and public IP address (15.206.183.145).

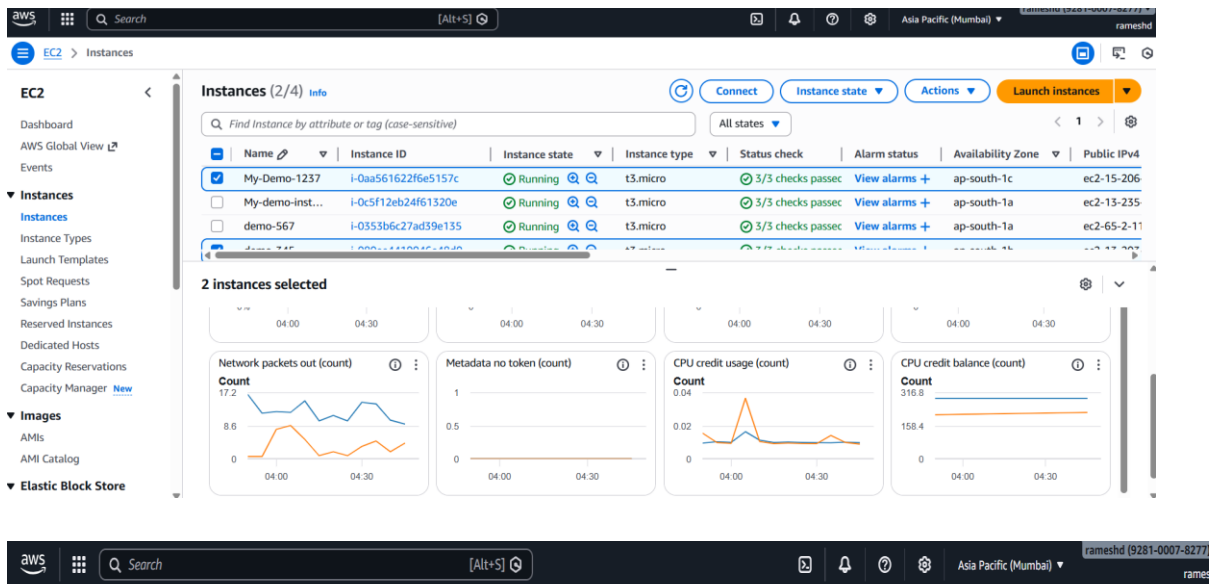
The screenshot shows the 'Connect to instance' page for the selected EC2 instance. The 'EC2 Instance Connect' tab is active, showing the instance ID, connection type (Public IP), and the public IP address (15.206.183.145). The 'Username' field is set to 'ubuntu'. A note at the bottom states: 'Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.'

Method 2: Using SSH (Linux/Mac/Windows)

1. Open terminal.
2. Navigate to folder containing .pem file.
3. Run:

```
chmod 400 MyKey.pem
```

```
ssh -i "MyKey.pem" ec2-user@
```
4. If successful, you get remote shell access to EC2.



```
https://ubuntu.com/aws/pro
Expanded Security Maintenance for Applications is not enabled.
78 updates can be applied immediately.
38 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-38-167:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-38-167:~$

i-0353b6c27ad39e135 (demo-567)
PublicIPs: 65.2.11.23 PrivateIPs: 172.31.38.167
```

```
78 updates can be applied immediately.
38 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
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applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-38-167:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-38-167:~$ ls
ubuntu@ip-172-31-38-167:~$ mkdir GIST
ubuntu@ip-172-31-38-167:~$ ls
GIST
ubuntu@ip-172-31-38-167:~$

i-0353b6c27ad39e135 (demo-567)
PublicIPs: 65.2.11.23 PrivateIPs: 172.31.38.167
```

Result:

- AWS account was successfully created.
- An EC2 instance was launched using t2.micro under Free Tier.
- The instance was successfully connected using EC2 Instance Connect and SSH client.

EXP NO:8

Cloud Storage and Database Services Using S3 (AWS), Blob (Azure), or GCP buckets and Cloud SQL/NoSQL

Aim

To create and manage cloud storage using Amazon S3 and to create a managed relational database instance using Amazon RDS (MySQL) and perform basic SQL operations.

Objectives

1. To understand cloud object storage using Amazon S3.
2. To perform file operations (Upload, Download, Delete) in S3.
3. To create a relational database using Amazon RDS.
4. To connect to RDS using MySQL client.
5. To perform SQL operations (CREATE, INSERT, SELECT, UPDATE, DELETE).

Tools / Software Required

- AWS Account
- Web Browser (Chrome / Edge)
- Internet Connection
- MySQL Client / MySQL Workbench
- Database Engine: MySQL

Amazon S3

Amazon S3 (Simple Storage Service) is an object storage service used to store and retrieve any amount of data from anywhere on the web. It stores data in **buckets** as **objects**.

Amazon RDS

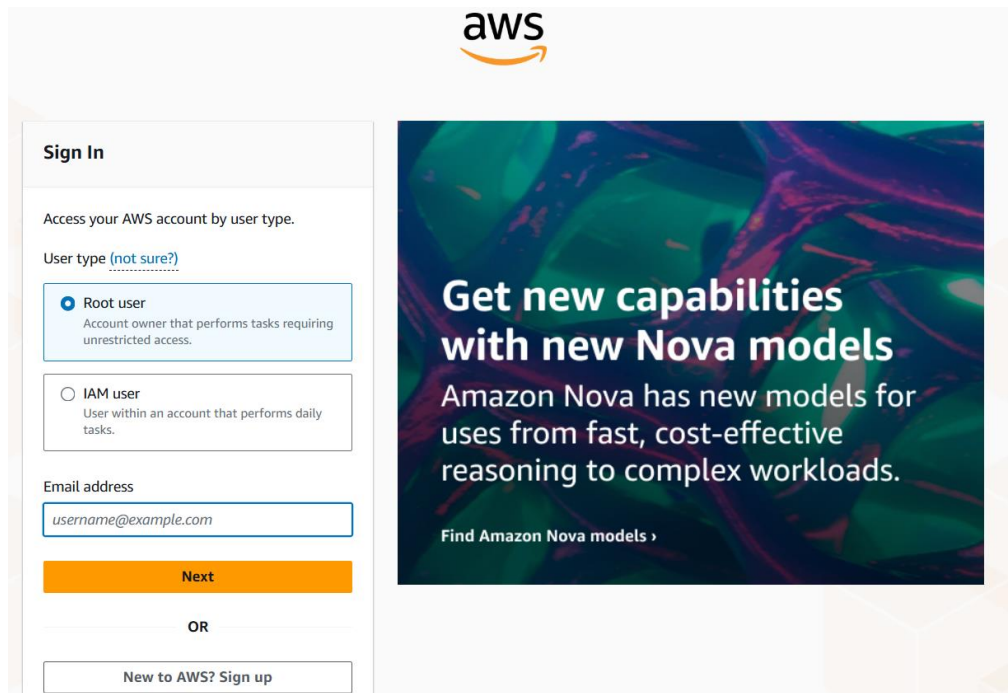
Amazon RDS (Relational Database Service) is a managed database service that supports engines like MySQL, PostgreSQL, MariaDB, etc. It automates backups, patching, and scaling.

Amazon S3

Step 1: Login to AWS Console

1. Open web browser
2. Go to: <https://aws.amazon.com/console>

3. Login using AWS credentials



The image shows the AWS Sign In page. At the top is the AWS logo. Below it, the 'Sign In' section is on the left, and a promotional banner for 'Get new capabilities with new Nova models' is on the right. The 'Sign In' section includes a link to 'Access your AWS account by user type.', a 'User type (not sure?)' link, two radio buttons for 'Root user' (selected) and 'IAM user', an 'Email address' field with 'username@example.com', a 'Next' button, an 'OR' separator, and a 'New to AWS? Sign up' button.

aws

Sign In

Access your AWS account by user type.

User type [\(not sure?\)](#)

☒ **Root user**
Account owner that performs tasks requiring unrestricted access.

☐ **IAM user**
User within an account that performs daily tasks.

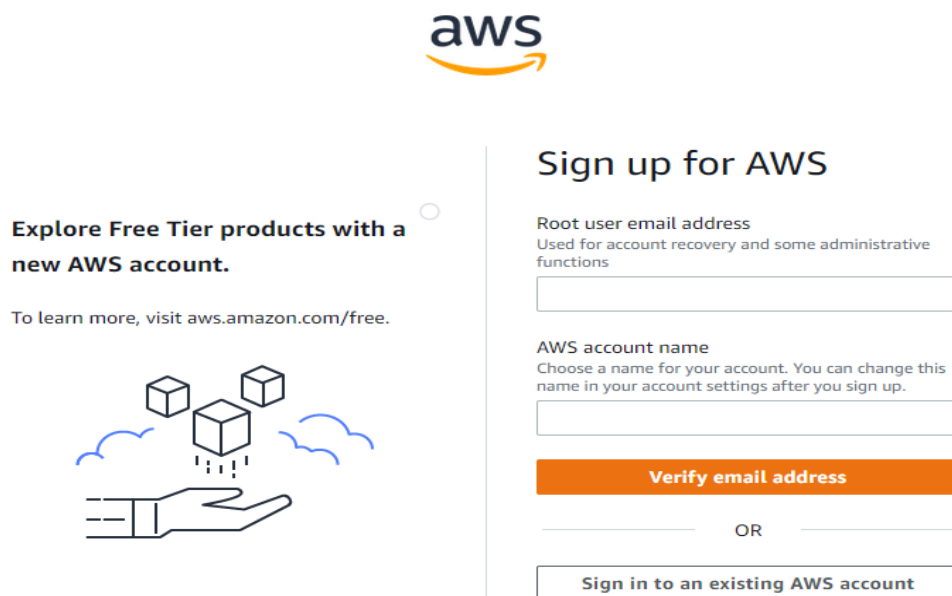
Email address

Next

OR

[New to AWS? Sign up](#)

Get new capabilities with new Nova models
Amazon Nova has new models for uses from fast, cost-effective reasoning to complex workloads.
[Find Amazon Nova models ›](#)



The image shows the AWS Sign up for AWS page. At the top is the AWS logo. Below it, the 'Sign up for AWS' section is on the right, and a promotional banner for 'Explore Free Tier products with a new AWS account.' is on the left. The 'Sign up for AWS' section includes a 'Root user email address' field, an 'AWS account name' field, a 'Verify email address' button, an 'OR' separator, and a 'Sign in to an existing AWS account' button.

aws

Sign up for AWS

Explore Free Tier products with a new AWS account.
To learn more, visit aws.amazon.com/free.

Root user email address
Used for account recovery and some administrative functions

AWS account name
Choose a name for your account. You can change this name in your account settings after you sign up.

Verify email address

OR

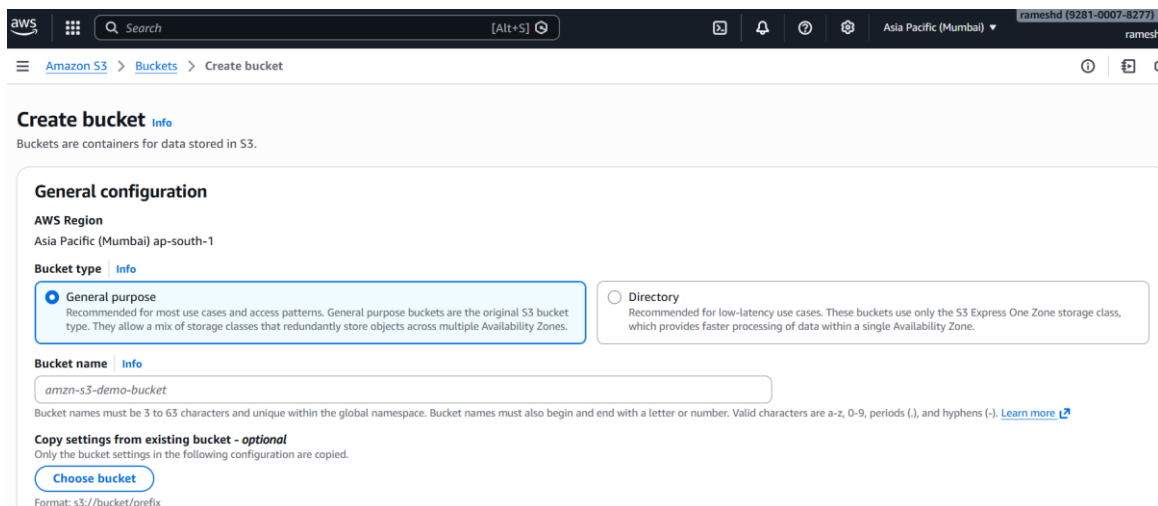
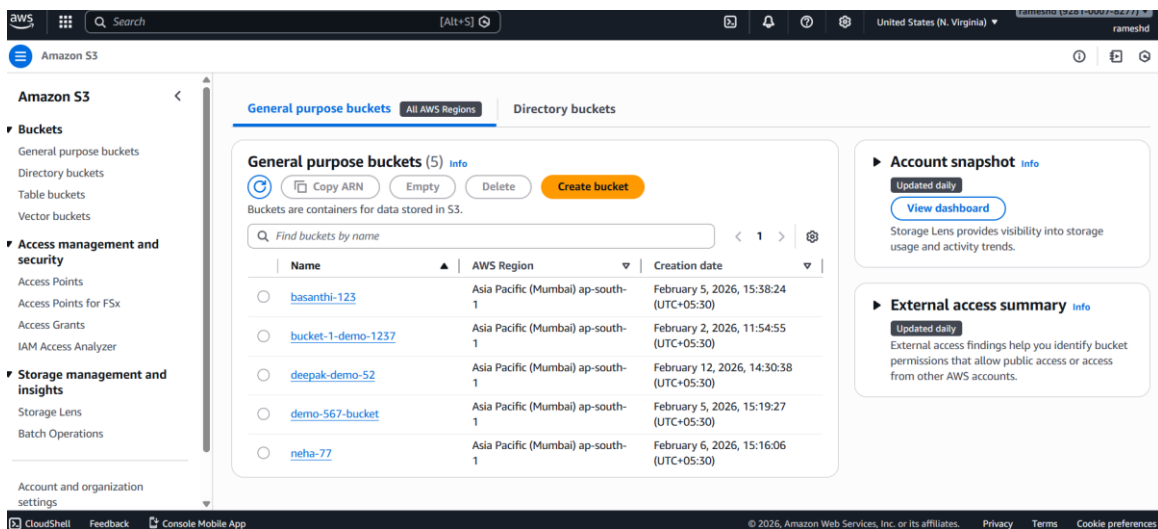
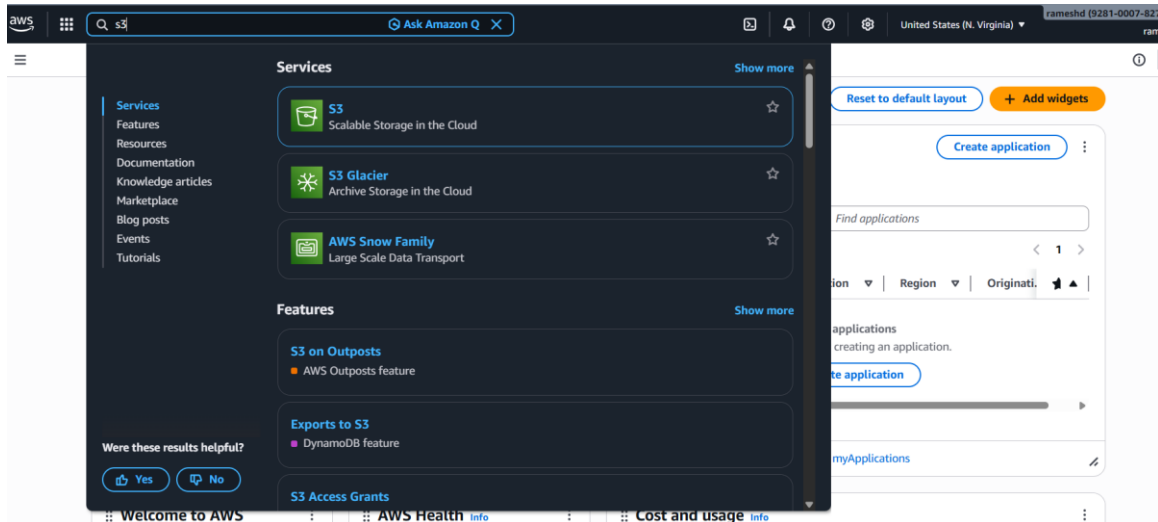
[Sign in to an existing AWS account](#)

Step 2: Open S3 Service

1. Search for S3
2. Click Amazon S3

Step 3: Create Bucket

- Click **Create Bucket**
- Bucket name: student-cloud-bucket
- Region: Select nearest region
- Object Ownership: ACLs disabled (Recommended)
- Block Public Access: Enabled
- Versioning: Disabled
- Click **Create bucket**



Step 4: Upload File

- Open bucket
- Click **Upload**
- Add files
- Select file
- Click **Upload**

Amazon S3

Buckets

demo-567-bucket

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

Storage management and insights

Storage Lens

Batch Operations

demo-567-bucket

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (0)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name

Type

Last modified

Size

Storage class

No objects

You don't have any objects in this bucket.

Upload

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

rameshd (9281-0007-8277)

rameshd

Amazon S3

Buckets

demo-567-bucket

Upload

Upload

Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (0)

Remove

Add files

Add folder

All files and folders in this table will be uploaded.

Find by name

Name

Folder

Type

Size

No files or folders

You have not chosen any files or folders to upload.

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

rameshd (9281-0007-8277)

rameshd

Amazon S3

Buckets

bucket-1-demo-1237

bucket-1-demo-1237

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name

Type

Last modified

Size

Storage class

Lab Daily Assessment (1).xlsx

xlsx

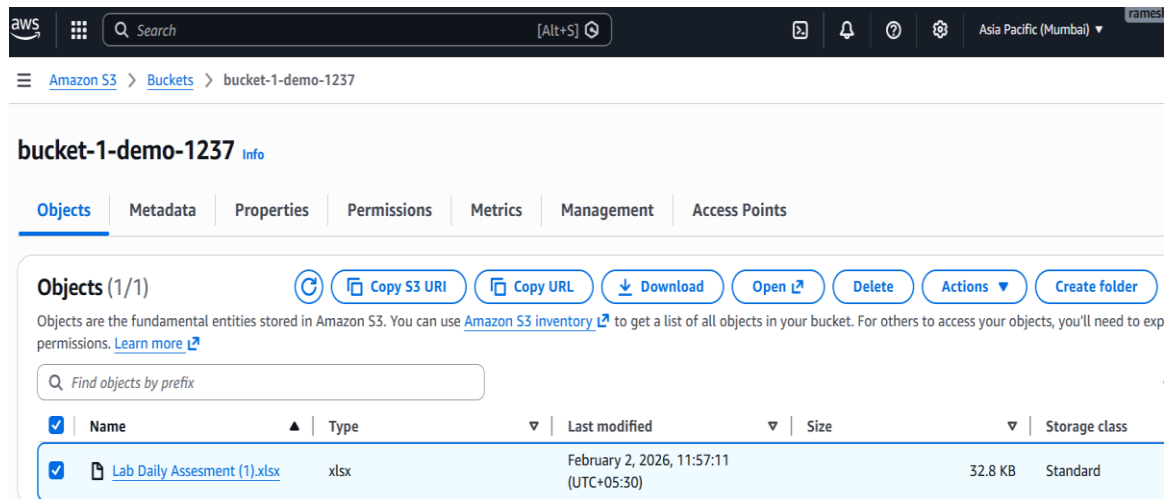
February 2, 2026, 11:57:11 (UTC+05:30)

32.8 KB

Standard

Step 5: Download File

- Select file
- Click **Download**



aws [Search] [Alt+S] Asia Pacific (Mumbai) ramesh

Amazon S3 > Buckets > bucket-1-demo-1237

bucket-1-demo-1237 [Info](#)

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [Access Points](#)

Objects (1/1) [Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#)

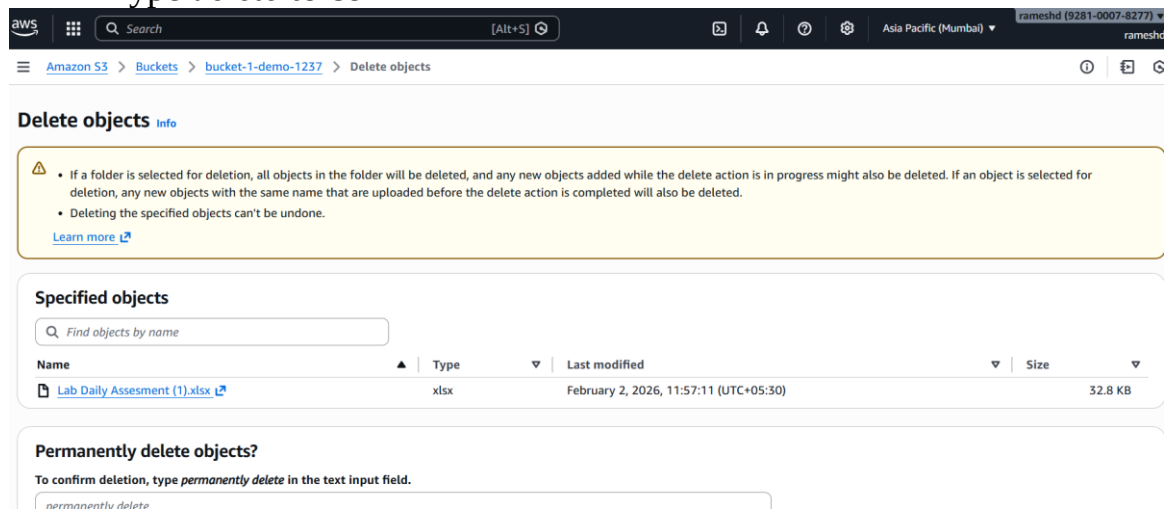
Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to exp permissions. [Learn more](#)

Find objects by prefix

☒	Name	Type	Last modified	Size	Storage class
☒	Lab Daily Assessment (1).xlsx	xlsx	February 2, 2026, 11:57:11 (UTC+05:30)	32.8 KB	Standard

Step 6: Delete File

- Select file
- Click **Delete**
- Type *delete* to confirm



aws [Search] [Alt+S] Asia Pacific (Mumbai) rameshd (9281-0007-8277) rameshd

Amazon S3 > Buckets > bucket-1-demo-1237 > Delete objects

Delete objects [Info](#)

Warning

- If a folder is selected for deletion, all objects in the folder will be deleted, and any new objects added while the delete action is in progress might also be deleted. If an object is selected for deletion, any new objects with the same name that are uploaded before the delete action is completed will also be deleted.
- Deleting the specified objects can't be undone.

[Learn more](#)

Specified objects

Find objects by name

Name	Type	Last modified	Size
Lab Daily Assessment (1).xlsx	xlsx	February 2, 2026, 11:57:11 (UTC+05:30)	32.8 KB

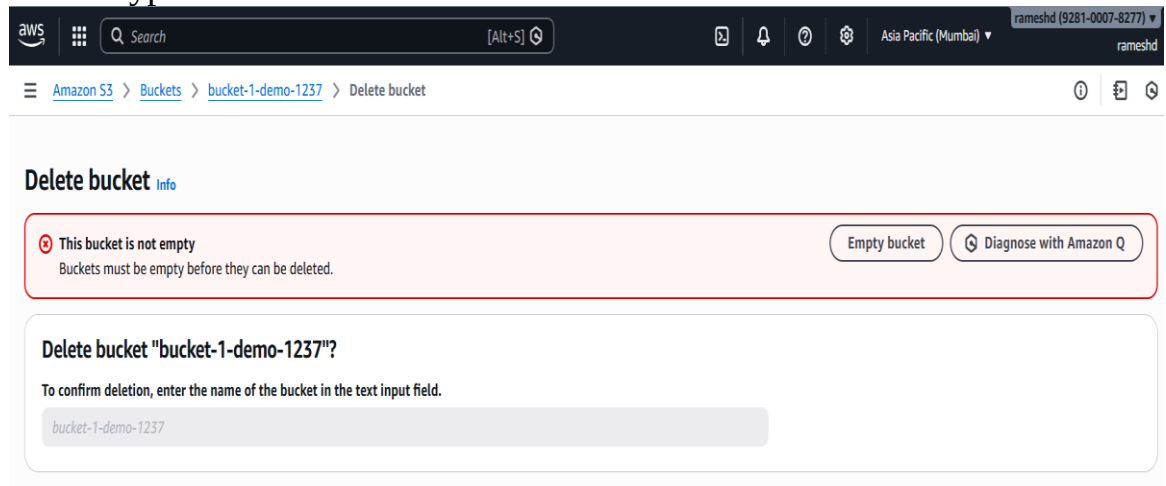
Permanently delete objects?

To confirm deletion, type *permanently delete* in the text input field.

permanently delete

Step 7: Delete Bucket

- Ensure bucket is empty
- Click **Delete**
- Type bucket name to confirm



aws [Search] [Alt+S] Asia Pacific (Mumbai) rameshd (9281-0007-8277) rameshd

Amazon S3 > Buckets > bucket-1-demo-1237 > Delete bucket

Delete bucket [Info](#)

Warning

This bucket is not empty
Buckets must be empty before they can be deleted.

[Empty bucket](#) [Diagnose with Amazon Q](#)

Delete bucket "bucket-1-demo-1237"?

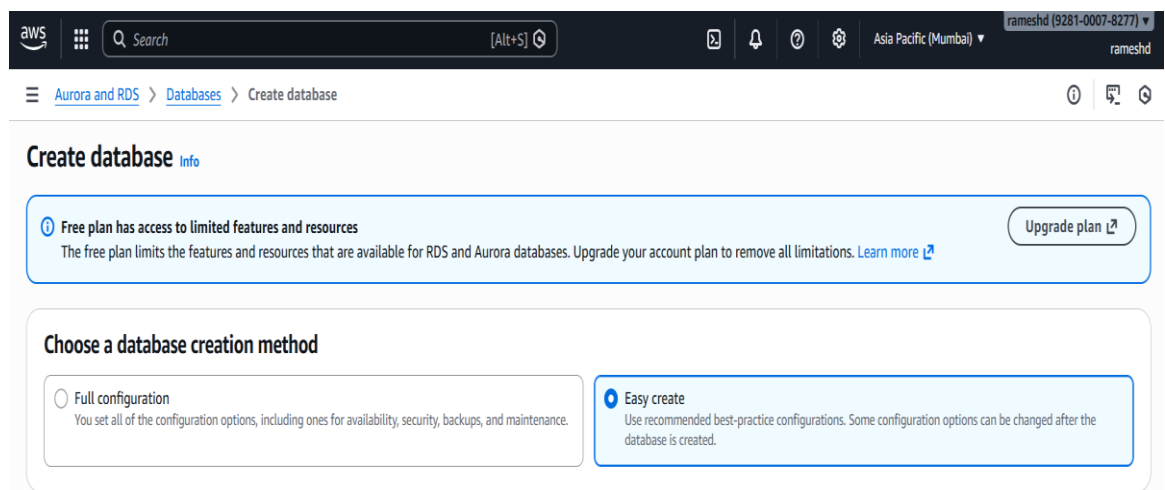
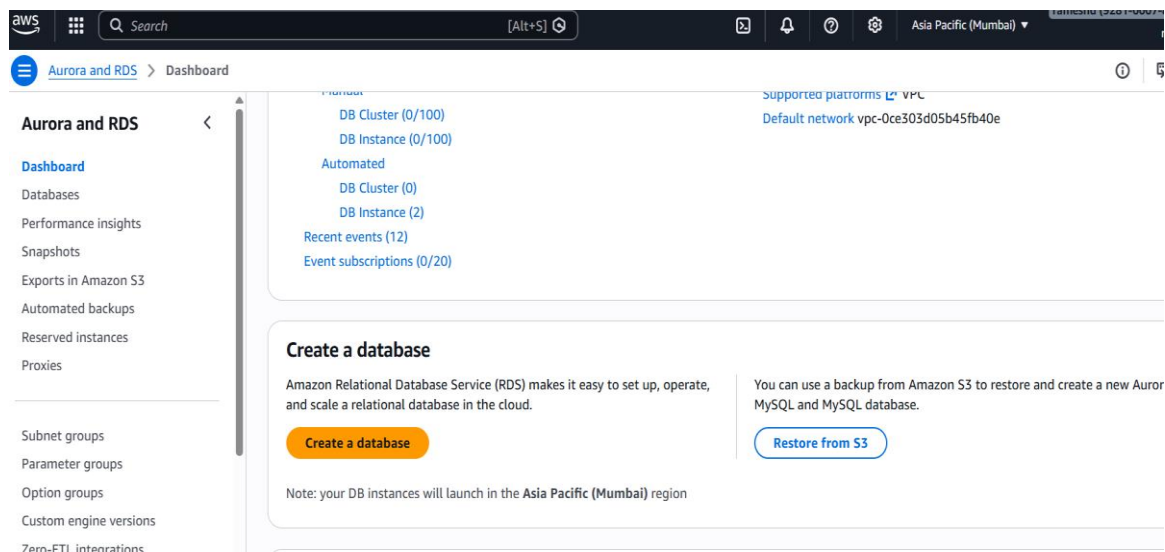
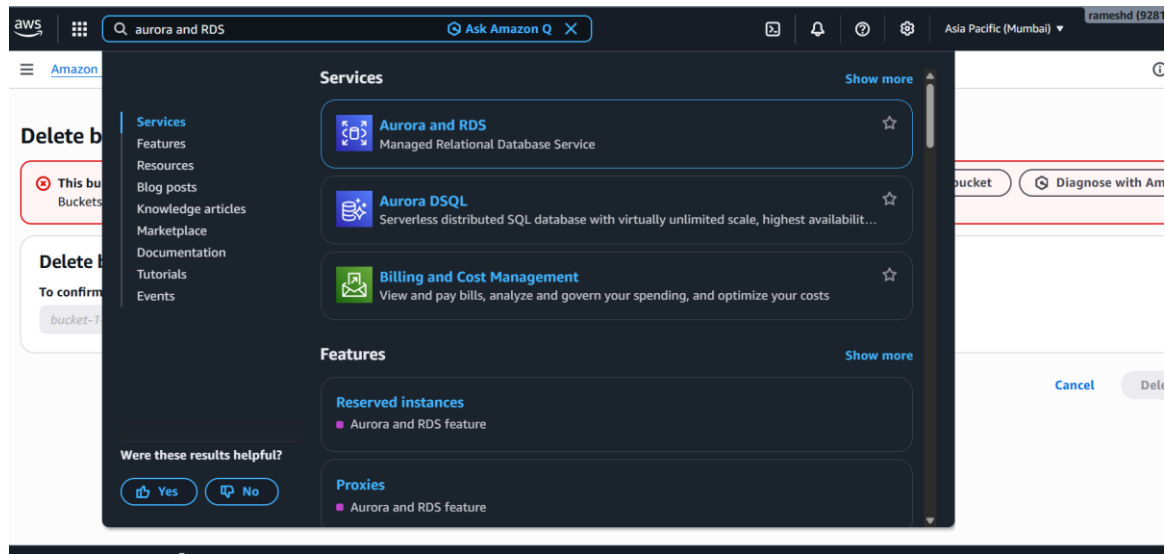
To confirm deletion, enter the name of the bucket in the text input field.

bucket-1-demo-1237

Procedure – Part B: Amazon RDS (MySQL)

Step 1: Open RDS Service

- Search for RDS
- Click **Create Database**



Step 2: Database Configuration

- Creation Method: Standard Create
- Engine: MySQL
- Template: Free Tier
- DB Instance Identifier: studentdb
- Master Username: admin
- Password: Admin@123
- Instance Class: db.t3.micro
- Storage: 20 GB (SSD)
- Public Access: Yes
- Initial DB Name: college
- Click **Create Database**

Wait until status becomes **Available**

The screenshot shows the AWS Management Console interface for creating a new database. The breadcrumb navigation indicates the path: Aurora and RDS > Databases > Create database. The 'Configuration' section is active, showing options for 'Engine type' and 'DB instance size'. Under 'Engine type', PostgreSQL is selected. Under 'DB instance size', the 'Free tier' option (db.t4g.micro) is selected. Below the configuration section, there are two input fields: one for the database name and another for confirming the master password. A note specifies that the password must be at least 8 printable ASCII characters and cannot contain certain symbols. Further down, there are two optional sections: 'Set up EC2 connection - optional' and 'View default settings for Easy create'. At the bottom right, there are two buttons: 'Cancel' and 'Create database'.

Configuration

Engine type [Info](#)

☐ Aurora (MySQL Compatible) ☐ Aurora (PostgreSQL Compatible) ☐ MySQL ☒ PostgreSQL

☐ MariaDB ☐ Oracle ☐ Microsoft SQL Server ☐ IBM Db2

DB instance size

☐ Production db.r7g.xlarge 4 vCPUs 37 GB RAM ☐ Dev/Test db.r7g.large 2 vCPUs 16 GB RAM ☒ Free tier db.t4g.micro 2 vCPUs 1 GB RAM

Database name

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Confirm master password [Info](#)

Set up EC2 connection - optional

You can also set up a connection to an EC2 instance after creating the database. Go to the database list page or the database details page, choose Actions, and then choose Set up to EC2 connection.

View default settings for Easy create

Easy create sets the following configurations to their default values, some of which can be changed later. If you want to change any of these settings now, use Standard create.

You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

[Cancel](#) [Create database](#)

aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) rameshd (9281-0007-8277) rameshd

Aurora and RDS > Databases

Aurora and RDS

- Dashboard
- Databases**
- Performance insights
- Snapshots
- Exports in Amazon S3
- Automated backups
- Reserved instances
- Proxies
- Subnet groups
- Parameter groups
- Option groups
- Custom engine versions
- Zero-ETL integrations

Creating database database-1

Your database might take a few minutes to launch. You can use settings from database-1 to simplify configuration of suggested database add-ons while we finish creating your DB for you.

[View connection details](#)

Databases (2) [Group resources](#) [Modify](#) [Actions](#) [Create database](#)

[Filter by databases](#)

DB identifier	Status	Role	Engine	Upgrade rollout order	Region
database-1	Creating	Instance	MySQL Co...	SECOND	-
demo-aimlb	Available	Instance	MySQL Co...	SECOND	ap-south-1b

aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) rameshd (9281-0007-8277) rameshd

Aurora and RDS > Databases > demo-aimlb

Aurora and RDS

- Dashboard
- Databases**
- Performance insights
- Snapshots
- Exports in Amazon S3
- Automated backups
- Reserved instances
- Proxies
- Subnet groups
- Parameter groups
- Option groups
- Custom engine versions
- Zero-ETL integrations

Creating database database-1

Your database might take a few minutes to launch. You can use settings from database-1 to simplify configuration of suggested database add-ons while we finish creating your DB for you.

[View connection details](#)

demo-aimlb [Modify](#) [Actions](#)

Summary

DB identifier demo-aimlb	Status Available	Role Instance	Engine MySQL Community	Recommendations
CPU 3.38%	Class db.t4g.micro	Current activity 0 Connections	Region & AZ ap-south-1b	

[Connectivity & security](#) [Monitoring](#) [Logs & events](#) [Configuration](#) [Zero-ETL integrations](#) [Maintenance & b](#)

Connect using [Info](#)

☒ **Code snippets**
Use when connecting through SDK, APIs, or third-party tools including agents.

☐ **CloudShell**
Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

☐ **Endpoints**
Use when connecting through any IDE interface.

Step 3: Get Endpoint

Example:

studentdb.cle2xyz.us-east-1.rds.amazonaws.com

aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) rameshd (9281-0007-8277) rameshd

Aurora and RDS > Databases > demo-aimlb

Aurora and RDS

- Dashboard
- Databases**
- Performance insights
- Snapshots
- Exports in Amazon S3
- Automated backups
- Reserved instances
- Proxies
- Subnet groups
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- Custom engine versions
- Zero-ETL integrations

Connect using [Info](#)

☐ **Code snippets**
Use when connecting through SDK, APIs, or third-party tools including agents.

☐ **CloudShell**
Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

☒ **Endpoints**
Use when connecting through any IDE interface.

Database name
mysql

Master username
admin

Internet access gateway
Disabled

Port
3306

Endpoint type
Instance endpoint

Additional configurations

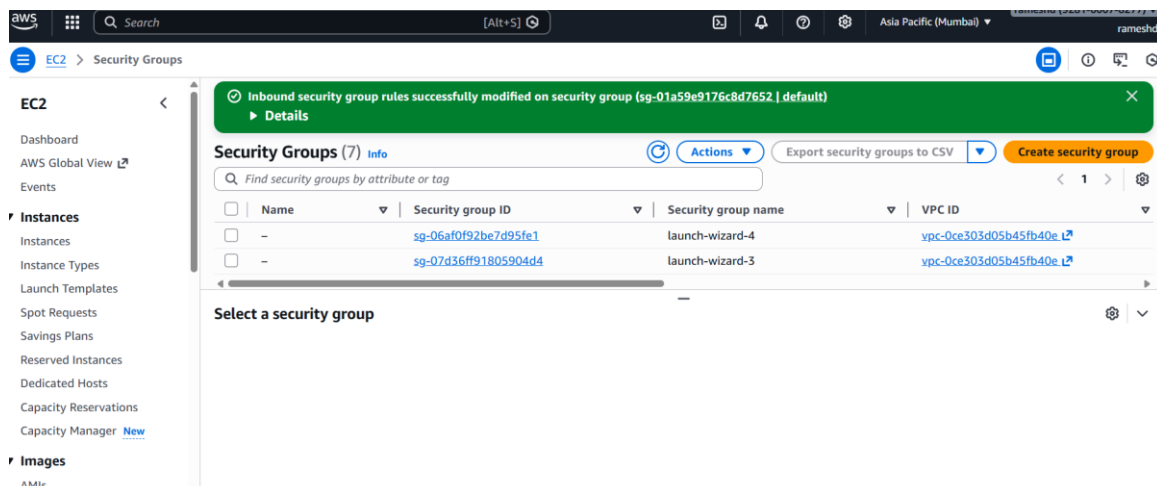
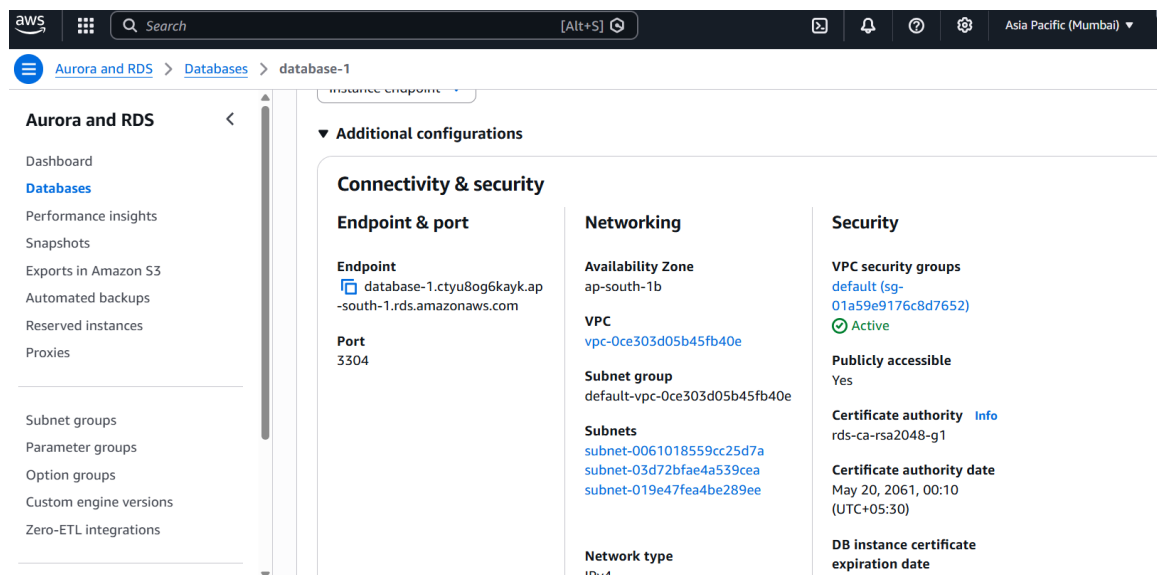
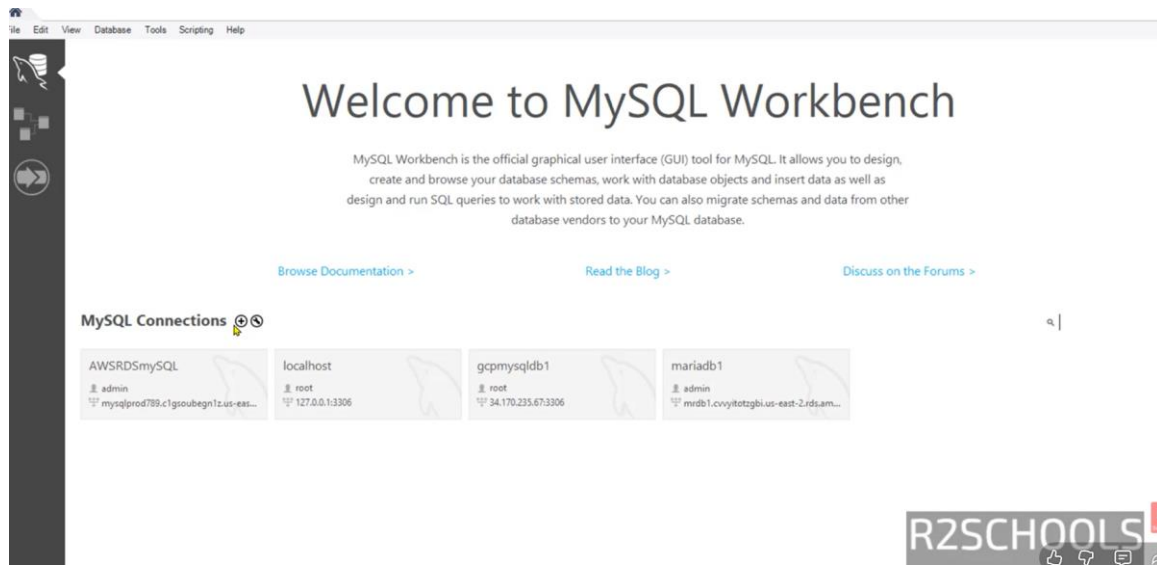
Connectivity & security

Endpoint & port	Networking	Security
Endpoint demo-aimlb.ctyu8og6kayk.a	Availability Zone ap-south-1b	VPC security groups default (sg-...)

SQL Operations

Connect to Database

```
mysql -h studentdb.cle2xyz.us-east-1.rds.amazonaws.com -u admin -p
```



Create Database

```
CREATE DATABASE labdb;
```

Use Database

```
USE labdb;
```

Create Table

```
CREATE TABLE students (  
    id INT PRIMARY KEY,  
    name VARCHAR(50),  
    department VARCHAR(30),  
    marks INT  
);
```

Insert Data

```
INSERT INTO students VALUES  
(1, 'Ramesh', 'CSE', 85),  
(2, 'Suresh', 'AI&ML', 90),  
(3, 'Mahesh', 'DS', 88);
```

Retrieve Data

```
SELECT * FROM students;
```

Update Data

```
UPDATE students SET marks = 95 WHERE id = 2;
```

Delete Data

```
DELETE FROM students WHERE id = 3;
```

Observations

Service	Operation	Status
Amazon S3	Bucket Creation	Successful
Amazon S3	File Upload	Successful
Amazon S3	File Download	Successful
Amazon S3	File Deletion	Successful
Amazon RDS DB Instance Creation		Successful
MySQL	Table Creation	Successful
MySQL	Insert / Select / Update / Delete	Successful

Output

- S3 bucket created successfully
- Files uploaded and downloaded successfully
- RDS MySQL instance created
- SQL operations executed successfully

Result

Thus, cloud storage services using Amazon S3 and managed database services using Amazon RDS were successfully implemented and verified.